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Solidification phases and their evolution during homogenization of a DC cast Al–8.35Zn–2.5Mg–2.25Cu alloy

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ABSTRACT

In the present work, the microstructure evolution of Al–8.35Zn–2.5Mg–2.25Cu alloy during homogenization treatments was investigated. The main grain boundary phases in Al–Zn–Mg–Cu alloy were $Mg(Zn,Cu,Al)_2$, which had the structure similar to $MgZn_2$ containing Al and Cu elements, and Al_7Cu_2Fe. The content of $Mg(Zn,Cu,Al)_2$ phase firstly decreased sharply within the initial 1 h, and then the mesh-like $Mg(Zn,Cu,Al)_2$ structure was gradually evolved to be discontinuous particles. The ratio of Al, Zn, Mg and Cu elements in $Mg(Zn,Cu,Al)_2$ phase was slightly affected by the homogenization treatment. Combining with other literature results, it could be concluded that the phase transition from $Mg(Zn,Al,Cu)_2$ to Al_2CuMg phases was very difficult when Zn content was higher than 8 wt.%. It is suggested that the significant effect of Zn might be explained by the diffusion behavior of Zn element.
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1. Introduction

Al–Zn–Mg–Cu alloys have been widely used in aerospace and military industries due to their high mechanical performance [1]. The mechanical properties of Al–Zn–Cu–Mg alloys are usually increased with the Zn amount [2]. However, more micro-segregation [3] and coarse intermetallic particles [4], (which are detrimental to the mechanical and corrosion properties [5,6]), will be formed in the Al alloy with high Zn content. It is well known that micro-segregation may be eliminated while large soluble non-equilibrium intermetallic phases can be re-dissolved during homogenization process [7]. Therefore, homogenization treatment is mandatory for the as-cast Al–Zn–Cu–Mg alloys to obtain a good service performance [8,9].

It has been reported that several intermetallic phases, such as η ($MgZn_2$) [2], T (AlZnMgCu) [7,10], and S (Al_2CuMg) [10], would be formed in Al–Zn–Mg–Cu alloys below the solidus temperature. Phase dissolution and transformation, which can be influenced by alloying elements and processing, will occur during homogenization [11]. Chen et al. [12] found that η phase was dissolved completely while T and S phases were still remained in 7055 alloy after 470 °C/50 h pretreatment. Meanwhile, only S phase was detected while other non-equilibrium eutectics were dissolved completely after 743 K/72 h homogenization treatment in Al–6.13Zn–2.65Mg–1.6Cu alloy [13]. Usually, S phase could be formed directly during solidification process [14,15]. Moreover, S phase could also be generated due to the transformation of AlZnMgCu phase during homogenization [13,16]. However, Zuo et al. [17]

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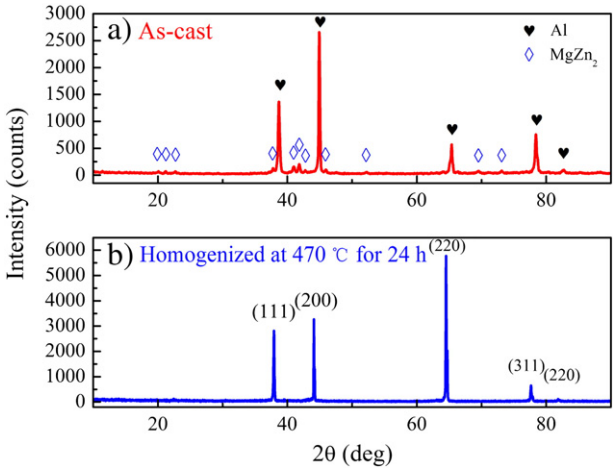

Fig. 1 – X-ray diffraction patterns of Al–Zn–Mg–Cu alloy (a) before and (b) after homogenized at 743 K/24.

reported that S phase was not detected in both as-cast and 743 K/24 h homogenized Al–9.97Zn–2.65Mg–1.94Cu alloy, and only η and T phases were found. Meanwhile, it was also found by Li et al. [18] that η and T phases had been dissolved completely and α-Al single-phase solid solution was formed after 1023 K/45 min homogenization treatment under high pressure (5 GPa).

Al–8.35Zn–2.5Mg–2.25Cu alloy demonstrates a very high mechanical strength (as high as about 700 MPa [19]). It shall be attributed to the high alloying element, especially Zn amount. For Al alloy with such high Zn content, homogenization treatment is mandatory before its processing and application. However, its phase evolution behavior during homogenization has not been well understood yet. Therefore, in the present work, the microstructure evolution of Al–8.35Zn–2.5Mg–2.25Cu alloy was investigated. Combining with the results from literatures, the effect of Zn amount on the S phase transformation was deeply discussed.

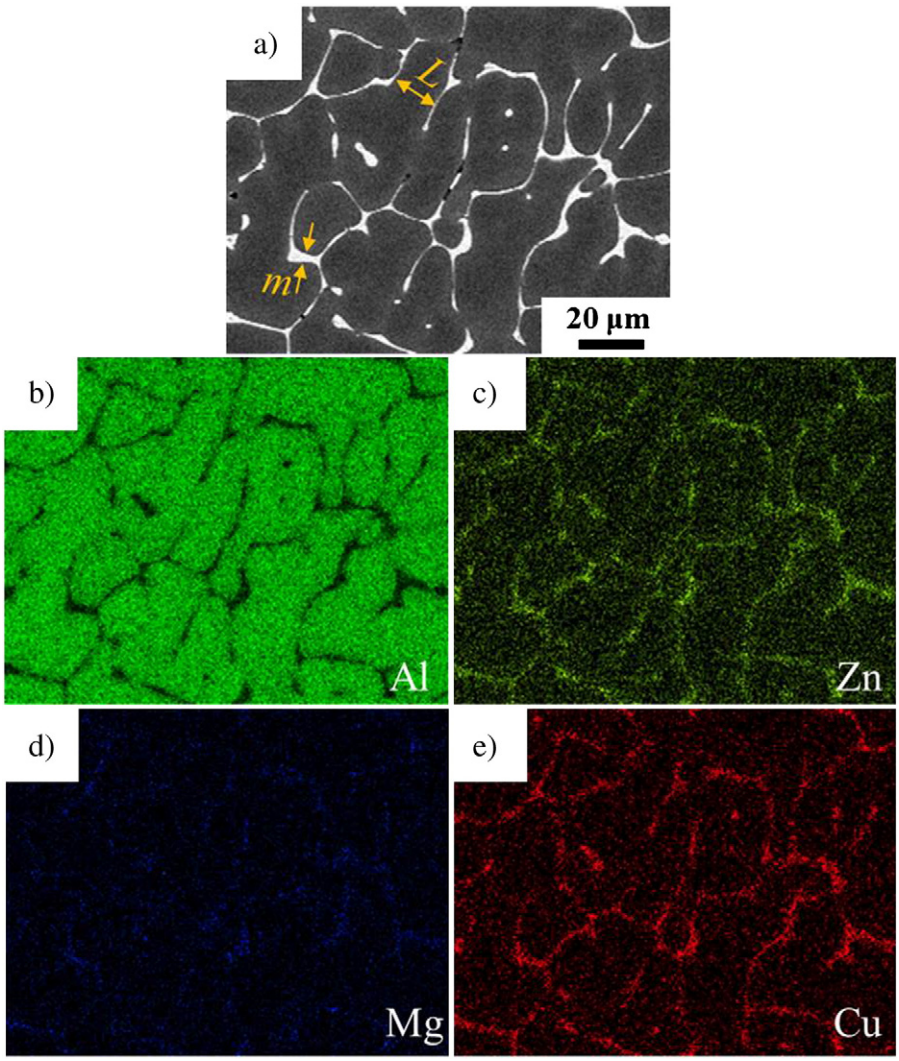

Fig. 2 – Microstructure and the main element distribution in the as-cast Al–Zn–Mg–Cu alloy. (a) Backscattered electron image, (b) Al, (c) Zn, (d) Mg, (e) Cu.

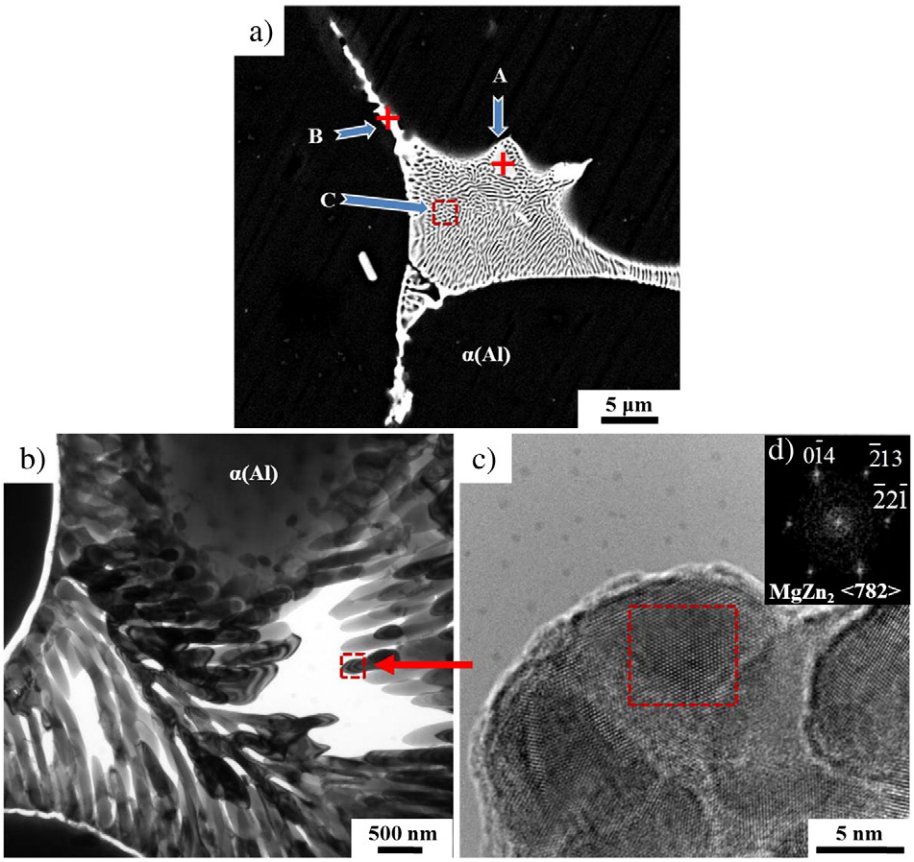

Fig. 3 – Microstructure of the as cast Al–Zn–Mg–Cu alloy. (a) SEM images of the as cast microstructure, (b) TEM image of AlZnMgCu phase, (c) HRTEM image of the eutectic in the red frame in (b), (d) FFT diffraction pattern of HRTEM image in the red frame in (c).

2. Experimental Detail

Al–Zn–Mg–Cu alloy was prepared by semi-continuous casting method. Its chemical composition was Al–8.35Zn–2.50Mg–2.25Cu–0.14Zr–0.04Si–0.08Fe–0.05Mn–0.04Cr–0.05Ni–0.03Ti (wt.%). The specimens with a dimension of 30 × 30 × 15 mm were used for homogenization treatments. The homogenization temperature (743 K in the present work) was determined by differential scanning calorimetry (DSC) analysis. The homogenization time was from 10 min to 36 h. The homogenization treatments were performed in a salt bath furnace with an accuracy of ±3 K.

DSC test was performed at a heating rate of 10 K/min to measure the melting points of the constituent phases. Phase identification was carried out by X-ray diffraction (XRD, Rigaku

Table 1 – Compound composition at the grain boundary of the as-cast alloy in Fig. 3(a) (mole fraction, %).

Item	Al	Zn	Mg	Cu	Fe	Zr	Phases
A	72.22	2.43	–	17.75	7.60	–	Al_7Cu_2Fe
B	35.43	21.46	32.82	10.30	–	–	AlZnMgCu phase
C	60.63	10.43	18.70	10.24	–	–	α (Al) + AlZnMgCu phase

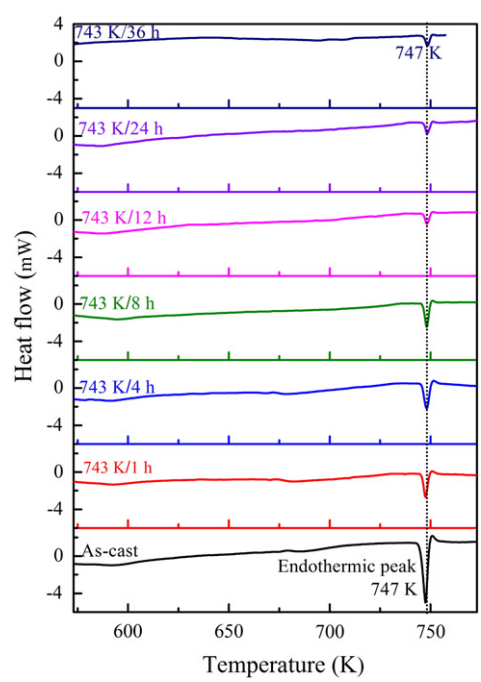

Fig. 4 – DSC curves of the Al–Zn–Mg–Cu alloy homogenized at 743 K for different times.

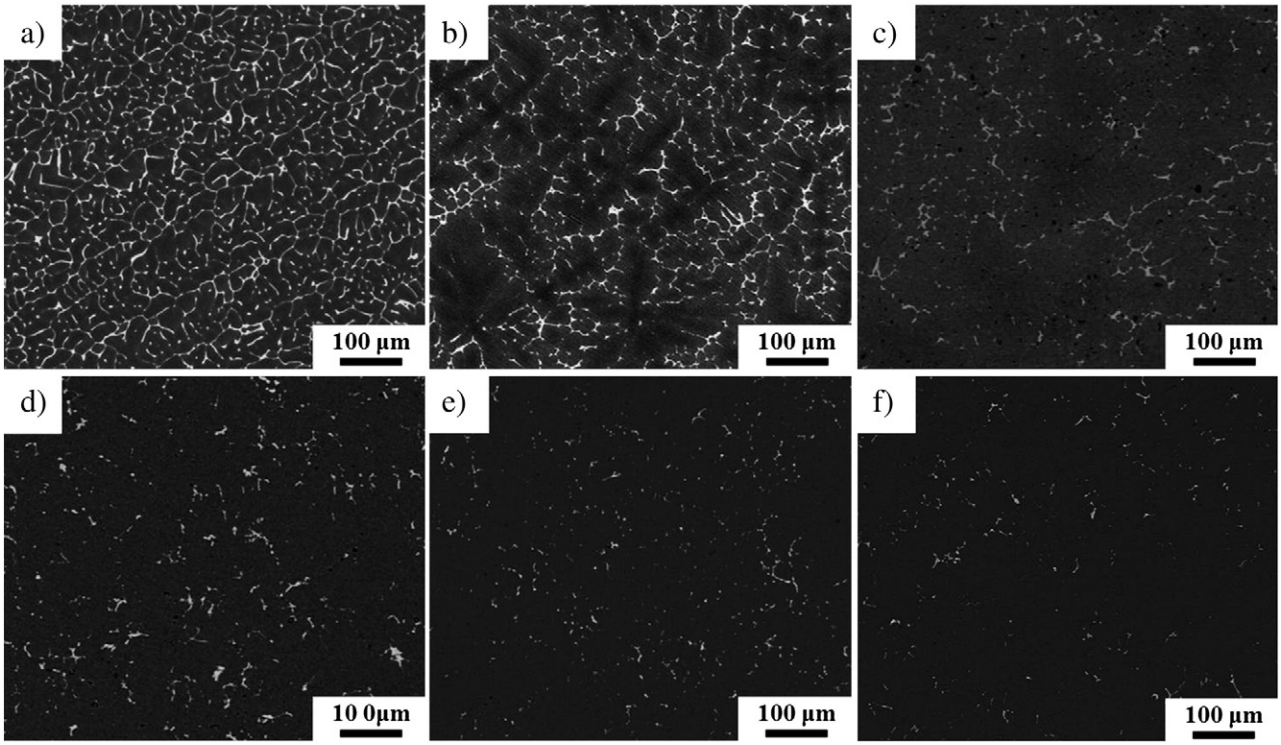

Fig. 5 – Microstructure evolution process of Al–Zn–Mg–Cu alloy during homogenization at 743 K. (a) As-cast, (b) 10 min, (c) 1 h, (d) 6 h, (e) 24 h, (f) 36 h.

D/max-2400) with CuKα radiation (0.15418 nm). Microstructure was investigated by FEI Quanta 200 scanning electron microscopy (SEM) equipped with energy dispersive spectroscopy (EDS). Further characterization of microstructure was conducted by a FEI TENCNAI G2 F30 transmission electron microscopy (TEM) equipment with an accelerated voltage of 120 kV and JEOL-2010CF high-resolution transmission electron microscopy (HRTEM) with an accelerated voltage of 200 kV.

3. Results and Discussion

3.1. Solidification Phases in DC cast Al–Zn–Mg–Cu alloy

X-ray diffraction patterns of Al–Zn–Mg–Cu alloy before homogenization treatment was shown in Fig. 1a. The main phases of as-cast Al–Zn–Mg–Cu alloy were α-Al and η ($MgZn_2$) phases. Representative microstructure of the as-cast Al–8.35Zn–2.5Mg–2.25Cu alloy was shown in Fig. 2a. Corresponding distribution of Al, Zn, Mg and Cu was shown in Fig. 2b–e, respectively. It was obvious that severe segregation of Zn, Mg and Cu elements along the Al grain boundaries was found. Lamellar or feathery phases were observed in high magnification (Fig. 3a). Few coarse particles (A point in Fig. 3a) embedded in the feathery phases were found. EDS analysis (Table 1) revealed that the composition of the coarse particle was close to Al_7Cu_2Fe [20] while white phases (B point in Fig. 3a) might be AlZnMgCu phase [21]. Only Al, Zn, Mg and Cu elements were detected by EDS at triangular grain boundary area (C area in Fig. 3a), suggesting that it might be the mixture of α-Al and AlZnMgCu phases.

Further TEM and HRTEM observations indicated that triangular grain boundary area was composed by α-Al (Fig. 3b) and feathery $MgZn_2$ phases (Fig. 3b and c). Generally, solid-solution Al and Cu elements were detected in $MgZn_2$ phase [17,18], which could explain the presence of Cu detected by EDS (Table 1). Therefore, it could be concluded that the triangular grain boundary area was composed by α-Al and feathery $MgZn_2$ phases containing dissolved Al and Cu elements.

DSC curves of as-cast Al–8.35Zn–2.5Mg–2.25Cu alloy was shown in Fig. 4. A clear endothermic peak at 747 K, which was related to the over-burnt of Al alloy, was observed. Therefore, the homogenization temperature should not exceed 747 K. Considering the accuracy of furnace, 743 K was chosen as the homogenization temperature in the present work.

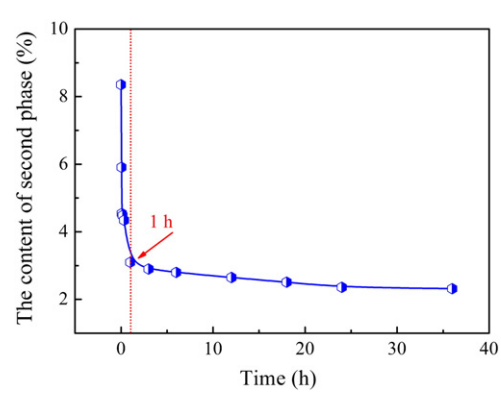

Fig. 6 – The statistical amount of the grain boundary phase after homogenization.

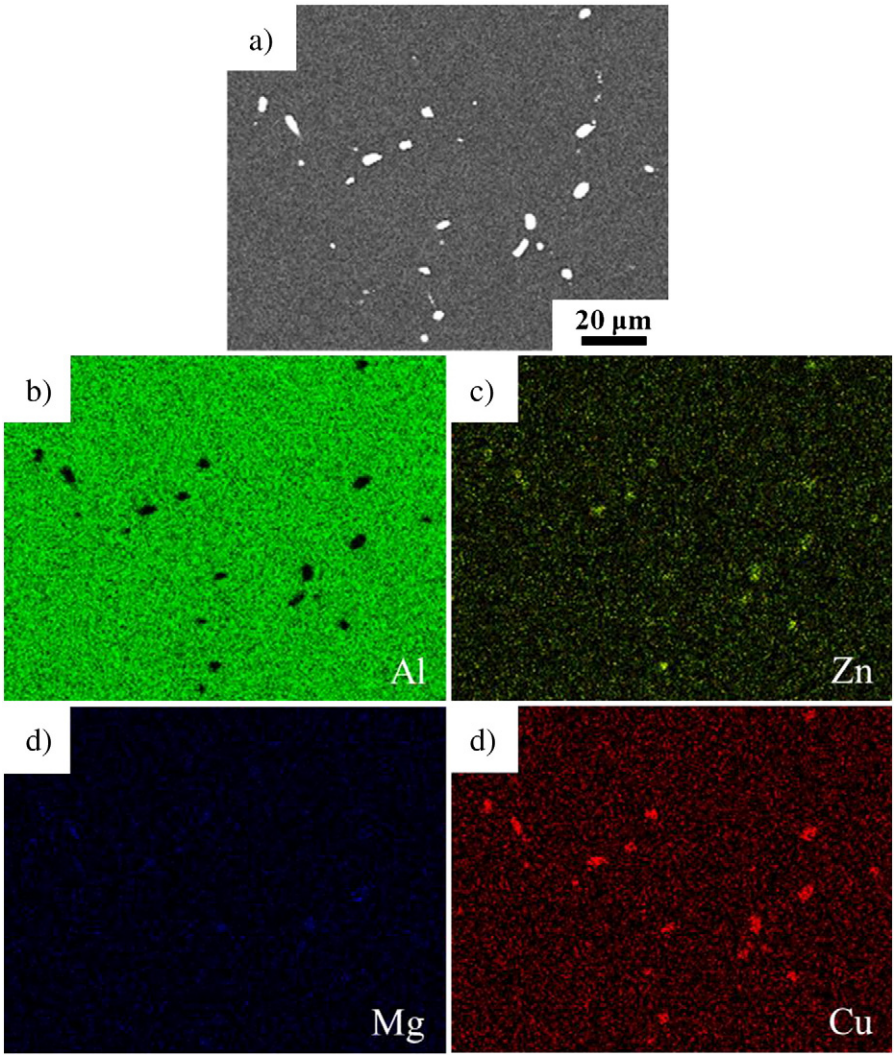

Fig. 7 – Microstructure and the main element distribution in the Al–Zn–Mg–Cu alloy after 743 K/24 h homogenization. (a) Backscattered electron image, (b) Al, (c) Zn, (d) Mg, (e) Cu.

3.2. Microstructure of Homogenized Alloy

DSC curves of Al–8.35Zn–2.5Mg–2.25Cu alloy homogenized at 743 K up to 36 h are shown in Fig. 4. Only one endothermic peak at 747 K, which is corresponding to the melting of AlZnMgCu phase, was observed regardless of homogenization time. This result was very similar to that of as-casting material. However, it should be noted that the value of endothermic peak was decreased with the increase of the homogenization time. Especially, the value of endothermic peak firstly decreased sharply within the initial 1 h. Then, it was decreased slightly with the further prolonging of homogenization time. Ultimately, they were affected slightly by the homogenization process when the homogenization time was increased from 12 to 36 h. These results imply that the content of $Mg(Zn,Cu,Al)_2$ phase firstly decreased sharply

Table 2 – EDS analysis of the constituent phases in Fig. 8 (mole fraction, %).

Item	Al	Zn	Mg	Cu	Fe	Phases
D	25.82	24.04	33.71	15.80	0.62	AlZnMgCu
E	73.69	1.60	2.94	15.05	6.72	Al_7Cu_2Fe

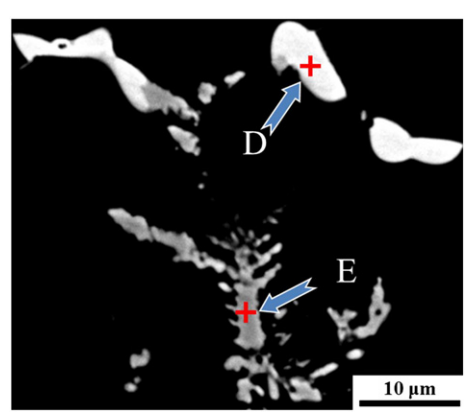

Fig. 8 – SEM backscattered images of the alloy homogenized at 743 K for 24 h.

within the initial 1 h and then it was decreased slightly with the further prolonging of homogenization time. But it still exists until the homogenization time increased to 36 h.

X-ray diffraction patterns of Al–Zn–Mg–Cu alloy after 743 K/24 h homogenization treatment were shown in Fig. 1b. Different to patterns of as-cast material (Fig. 1a), only peaks of α-Al were detected. It indicated that $MgZn_2$ phase, which was formed during solidification process, had been almost dissolved into Al matrix. Microstructure evolution process of Al–Zn–Mg–Cu alloy during homogenization was shown in Fig. 5. High amount of mesh AlZnMgCu phase along Al grain boundary was still observed after 10 min of homogenization (Fig. 5b). With the prolonging of homogenization time, the amount of the grain boundary phase decreased significantly and the mesh-like structure evolved to be discontinuous particles. The fraction of residual phase was determined by using the Image tool software to analyze at least five backscattered SEM images of each status. The statistical amount of the grain boundary phase after homogenization was shown in Fig. 6. The content of the grain boundary phase firstly decreased sharply within the initial 1 h. Then, its amount decreased slightly with the further prolonging of homogenization time.

Element distribution of Al, Zn, Mg and Cu after 743 K/24 h homogenization was shown in Fig. 7. Compared with the as-casting material, the homogenized alloy showed much low segregation characterization although very few phase at grain boundary could still be observed. Alloying element, such as Cu, was distributed much more uniform within the grain. Two types of residual particles were observed. As listed in Table 2, the white particles (marked as D in Fig. 8) and the gray fish-bone phases (marked as E in Fig. 8) were AlZnMgCu and Al_7Cu_2Fe phases, respectively. The variations of time and composition of elements Al, Zn, Mg and Cu in AlZnMgCu phase during homogenization at 743 K were shown in Fig. 9a–d, respectively. The ratio of Al, Zn, Mg and Cu elements in AlZnMgCu phase was slightly affected by the homogenization treatment.

Microstructure of 743 K/24 h homogenized Al–Zn–Mg–Cu alloy was further investigated by TEM, as shown in Fig. 10. The residual particles were mainly distributed along the grain boundary (Fig. 10a and d). The composition of the coarse phase (marked as F in Fig. 10a) was Al, Zn, Mg and Cu elements. However, electron diffraction pattern (Fig. 10c) indicated that the coarse phase had the structure similar to $MgZn_2$ phase. It had been widely reported that solid-solution Al and Cu elements were detected in $MgZn_2$ phase [17,18]. It could be concluded that the AlZnMgCu phase, which was formed in the casting process, was $MgZn_2$ dissolved with Al and Cu elements. Thus, it could be written as $Mg(Zn,Al,Cu)_2$. Moreover, compared with microstructure and EDS analysis results before and after homogenization treatment, it could be concluded that the structure and the composition of $Mg(Zn,Al,Cu)_2$ phase in the present work were slightly affected by the homogenization process. The residual particle marked by G in Fig. 10d was identified as Al_7Cu_2Fe phase based on the EDS result (Fig. 10e) and the diffraction pattern (Fig.10f).

3.3. Discussion

3.3.1. The Influence of AlZnMgCu Phase Distribution of the Alloy Homogenization Process

As mentioned in Figs. 2 and 7, the main phase that dissolved during homogenization was $Mg(Zn,Al,Cu)_2$ [17,18]. Usually,

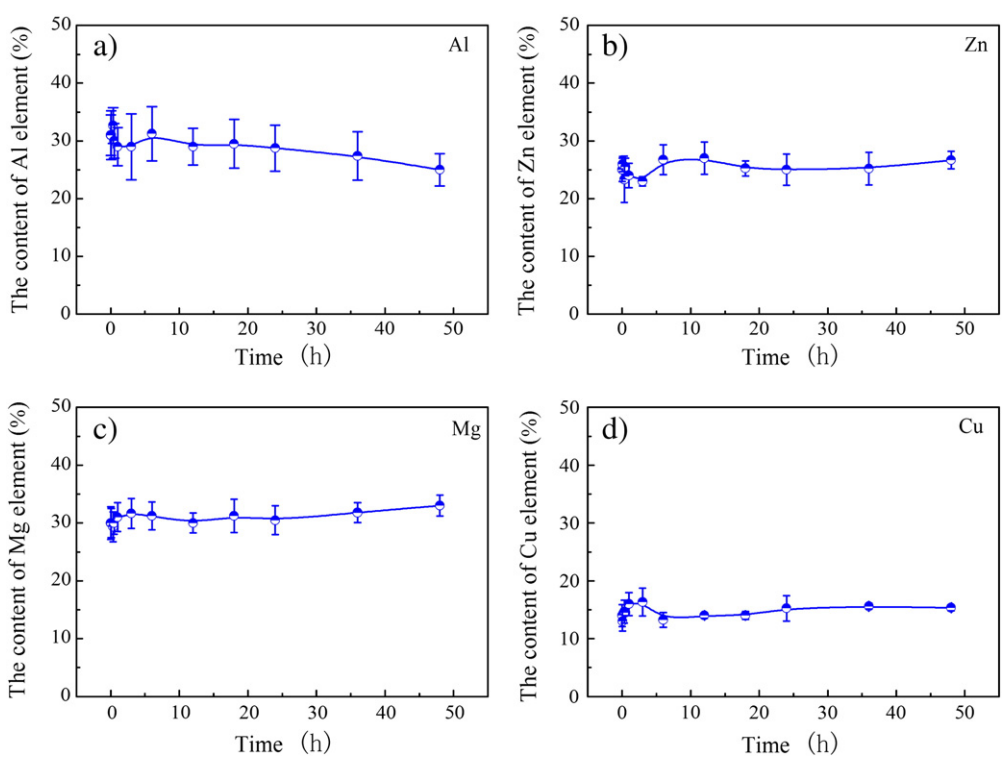

Fig. 9 – The variations of time and composition of elements (a) Al, (b) Zn, (c) Mg and (d) Cu in AlZnMgCu phase during homogenization at 743 K.

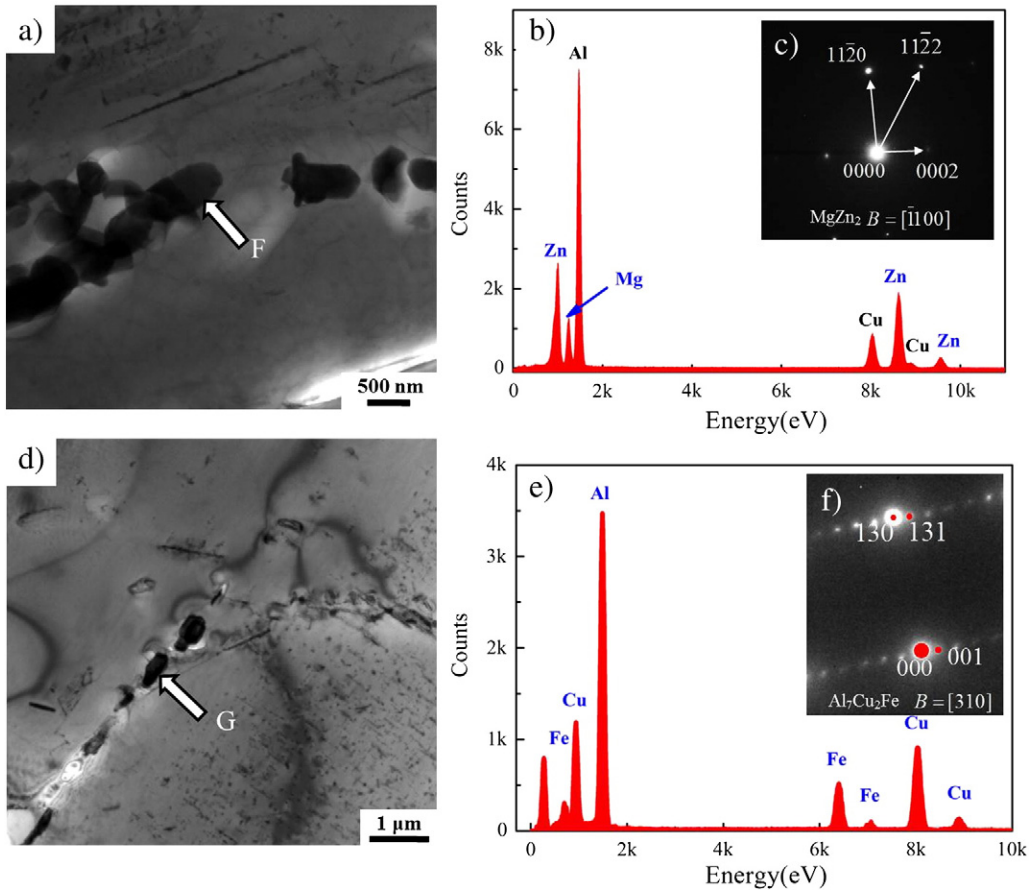

Fig. 10 – TEM graphs and diffraction pattern of the coarse phases after 743 K/24 h homogenization. (a) The morphology of $Mg(Al,Zn,Cu)_2$ phase, (b) EDS analysis of the area marked by F, (c) diffraction pattern of the area marked by F, (d) the morphology of Al_7Cu_2Fe, (e) EDS analysis of the area marked by G, (f) diffraction pattern of the area marked by G.

homogenization time was determined by the spacing and thickness of $Mg(Zn,Al,Cu)_2$. Effect of the spacing (L) on the homogenization process would be discussed by the homogenization kinetic analysis.

The line scanning analysis of the as-cast Al–Zn–Mg–Cu alloy was shown in Fig. 11. Obviously, the distribution of Zn, Mg and Cu elements along interdendritic region varied periodically. Thus, it was feasible to use Fourier series components in a cosine function to approach the initial concentration of the elements along the interdendritic region [22].

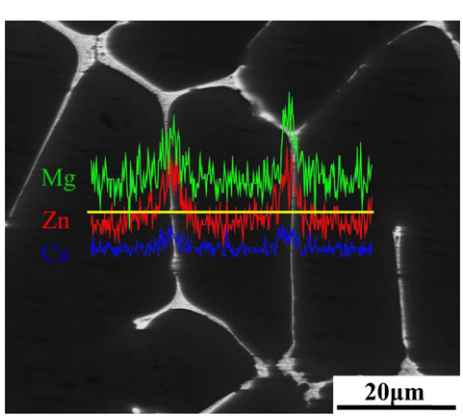

Fig. 11 – Line scanning analysis of as-cast Al–Zn–Mg–Cu alloy.

$$c(x) = \bar{c} + \left(\frac{1}{2}\Delta c_0\right)\cos\frac{2\pi x}{L} \tag{1}$$

where $\bar{c}$ was the average concentration of the element, L was the interdendritic spacing and Δc_0 was the concentration difference between the grain boundary and grain. According to the second Fick's law and the boundary conditions, $c(x)$ was given as [23]:

$$c(x) = \frac{1}{2}\Delta c_0 \exp\left(-\frac{4\pi^2}{L^2}Dt\right). \tag{2}$$

When the concentration difference of alloying elements was decreased to $\delta \times (1/2)\Delta c_0$, then:

$$\delta \times \frac{1}{2}\Delta c_0 = \frac{1}{2}\Delta c_0 \exp\left(-\frac{4\pi^2}{L^2}Dt\right). \tag{3}$$

By substituting

$$D = D_0 \exp\left(-\frac{Q}{RT}\right) \tag{4}$$

into Eq. (3), the equation could be rewritten as:

$$\frac{1}{T} = \frac{R}{Q}\ln(t) - \frac{R}{Q}\ln\left(\frac{L^2 \ln\delta}{4\pi^2 D_0}\right) \tag{5}$$

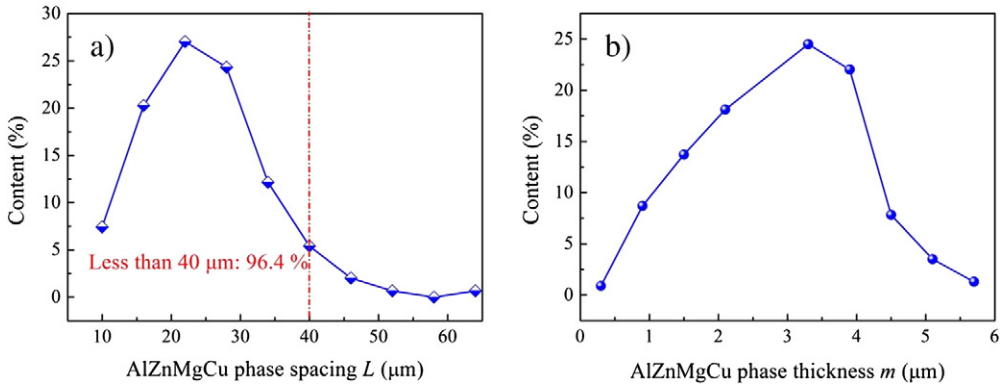

Fig. 12 – The spacing (a) and thickness (b) of the AlZnMgCu phase of the as-cast Al–Zn–Mg–Cu alloy.

where T ant t were the homogenization temperature and time, respectively; and R, D_0 and Q were the gas constant (8.314 J/(mol·K)), frequency factor and activation energy barrier, respectively.

Assuming the element distribution was homogeneous when the composition segregation amplitude reduced to 1%, i.e.: δ = 1% [24]. Then the corresponding equation to describe the homogenization kinetic could be expressed as:

$$\frac{1}{T} = \frac{R}{Q} \ln\left(\frac{4\pi^2 D_0 t}{4.6L^2}\right). \tag{6}$$

In the present work, as marked in Fig. 2a, the spacing (L) and thickness (m) of AlZnMgCu phase were used to describe their distribution in the as-cast alloy. More than 96% of spacing L was less than 40 μm while the maximum L was about 65 μm (Fig. 12a). Most of AlZnMgCu thickness m was laid between 2 and 4 μm while the maximum m was about 5.7 μm (Fig. 12b).

For a given alloy, the value of D_0 and Q could be obtained; hence the corresponding kinetic curve could be simulated. It was reported that the diffusion coefficient of Cu element in Al was much lower than that of Mg and Zn at the same temperature [20,21]. From the element distribution results in Fig. 2, similar qualitative conclusion could also be drawn. As a result, the homogenization was mainly affected by the diffusion process of Cu. Substituting D_0(Cu) (0.084 cm^2/s) and Q(Cu) (136.8 KJ/mol) into Eq. (6), the homogenization kinetic curve of AlZnMgCu phase with different spacings was obtained, as shown in Fig. 13. It was obvious that with the increase of AlZnMgCu phase spacing, higher temperature and longer time were needed for fully homogenization. Moreover, temperature demonstrated more significant effect on the homogenization process than time. However, according to the result of DSC analysis, the homogenization temperature should not exceed 474 °C. The homogenization time for AlZnMgCu phase with different spacings at 743 K was shown in Fig. 14. The homogenization time increased sharply with the spacing L. The homogenization time for AlZnMgCu phase with spacings of 40 and 65 μm were 26 and 68 h, respectively. The line scanning analysis of Al–Zn–Mg–Cu alloy homogenized at 743 K for 24 h was shown in Fig. 15. It indicated that, after homogenization, the segregation of the main elements Zn, Mg and Cu along the Al grain boundaries were almost eliminated and the element distribution was homogeneous. As mentioned above, the spacing of more than 96% AlZnMgCu phase was less than 40 μm, and the time required to complete the homogenization was about 26 h. It was consistent with the results of the experimental in the present work.

Effect of dendritic thickness (m) on the homogenization time (τ_s) could be expressed as in Eq. (7):

$$\tau_s = am^b \tag{7}$$

where a and b were the constants of alloys.

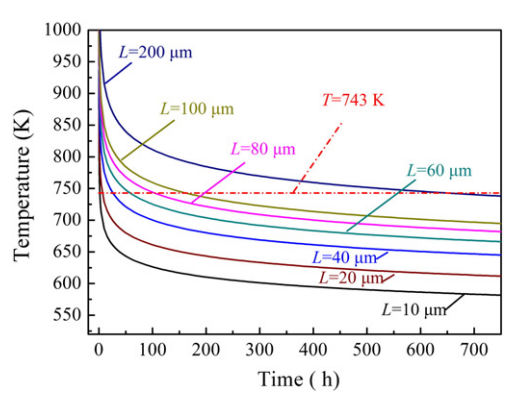

Fig. 13 – The homogenization kinetics curves of the alloy for different AlZnMgCu phase spacings.

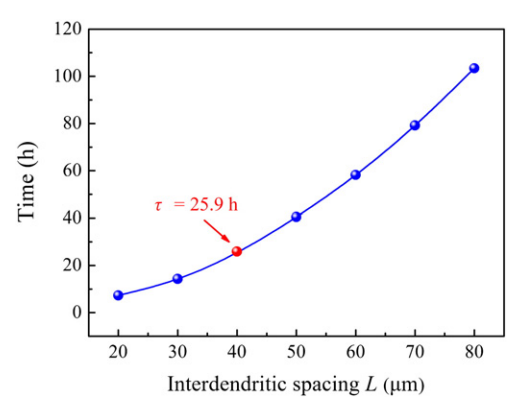

Fig. 14 – The homogenization time for AlZnMgCu phase with different spacings at 743 K.

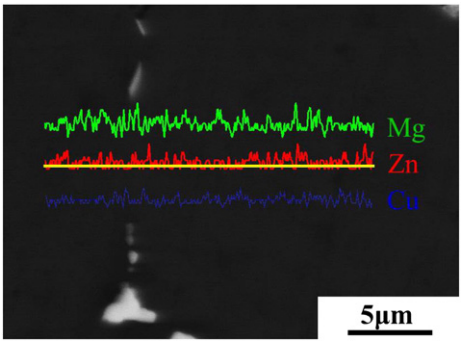

Fig. 15 – Line scanning analysis of Al–Zn–Mg–Cu alloy homogenized at 743 K for 24 h.

From Eq. (7), it could be seen that a little increment in dendritic thickness (m) would lead to a significant increase of time for complete homogenization. The dendritic thickness in the present work was from 0.3 to 5.5 μm. In Al alloy, the value of b was usually between 1.5 and 2.5 [25]. The dissolution time for 5.5 μm particles would be about 300 times than that for 0.3 μm particles if $b = 2$ was adopted. Therefore, these few dendrites with large thickness would be very difficult to be fully eliminated. Therefore, a small amount of AlZnMgCu phases still existed until the homogenization time increased to 36 h.

3.3.2. *The Influence of Zn Content on the Alloy Homogenization Process*

Fan et al. [16] found that the $Mg(Zn,Al,Cu)_2$ phase with relative low melting point in Al–6.31Zn–2.33Mg–1.7Cu aluminum alloys was transformed to S phase (Al_2CuMg) with high melting point in the initial stage of homogenization. Therefore, high amount of S phase was observed in their work. However, very few S phases were observed in the present work. According to the Al–Zn–Mg–Cu quaternary system phase diagram [25], η ($MgZn_2$) phase, S phase, T (Al_6CuMg_4) phase and T ($Al_2Mg_3Zn_3$) phase could be formed by Al, Zn, Mg and Cu elements. Meanwhile, T (Al_6CuMg_4) phase and T ($Al_2Mg_3Zn_3$) phase were usually named as T (AlZnMgCu) since they were infinite, mutual soluble. It had been concluded that S phase in semi-continuous casting Al–Zn–Mg–Cu alloy could only be formed when the amount of Zn, Cu, and Mg was higher than 3, 1 and 1 wt.%, respectively. Moreover, T (AlZnMgCu) or $Mg(Zn,Al,Cu)_2$ would be formed when the ratio of Zn/Mg ratio was lower or higher than 2.2, respectively. The Zn/Mg ratio was about 3.3 in the present work. Thus, the main phases of as-cast materials were α-Al and $Mg(Zn,Al,Cu)_2$.

Effect of Cu/Mg ratio and Zn content on the phase transition from $Mg(Zn,Al,Cu)_2$ to S (Al_2CuMg) phases during homogenization were summarized in Fig. 16. It was obvious that the phase transition behavior was significantly affected by Zn amount. The phase transition from $Mg(Zn,Al,Cu)_2$ to Al_2CuMg phases was rarely reported when Zn content was higher than 8 wt.%. This trend was not summarized in the past. Compared with their composition, it seemed that the transition from $Mg(Zn,Cu,Al)_2$ to Al_2CuMg phases would occur if Zn was dissolved into Al matrix. Therefore, it was

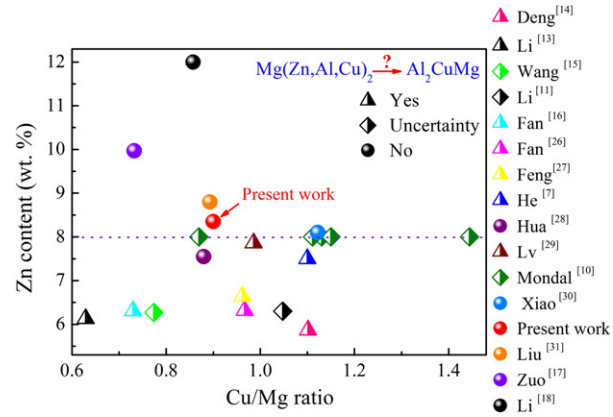

Fig. 16 – Effect of Cu/Mg ratio and Zn content on the phase transition from $Mg(Zn,Al,Cu)_2$ to S (Al_2CuMg) phases during homogenization [7,10,11,13,14,15,16,17,18,26,27,28,29,30,31].

suggested that the significant effect of Zn might be explained by the diffusion behavior of Zn element.

The diffusion of the solute within the grain during homogenization was analyzed by EDS. The results were shown in Fig. 17. The full line gave the solute distribution data of Al–8.35Zn–2.5Mg–2.25Cu alloy in the present paper. The dotted line was the solute distribution data of Al–6.32Zn–2.5Mg–2.25Cu alloy in Fan's research [32]. The element composition of the two alloys was very close except the Zn content had a 2% difference. It could be seen from Fig. 17 that the Zn distribution in the two as cast alloys was close. With the prolonging of the homogenization, part of Zn diffused to the Al matrix. As the Zn content was different in the two alloys, the Zn content in the Al matrix of Al–8.35Zn–2.5Mg–2.25Cu was 2% higher than that in Al–6.32Zn–2.4Mg–2.32Cu.

According to Fick's first law of diffusion:

$$J = -D\frac{d\rho}{dx}. \tag{8}$$

The diffusive flux (J) was proportional to the concentration gradient in x dimension. The amount of $Mg(Zn,Cu,Al)_2$ phase increased with Zn amount due to solid solubility low

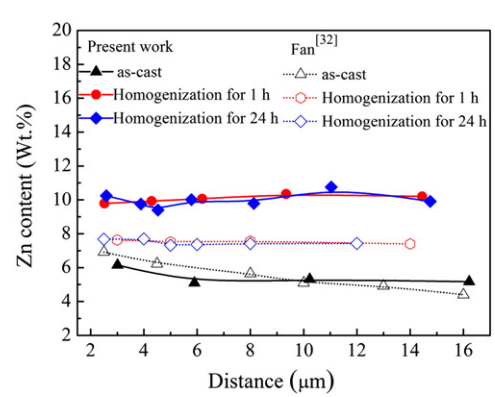

Fig. 17 – Evolution of Zn element distribution during homogenization.

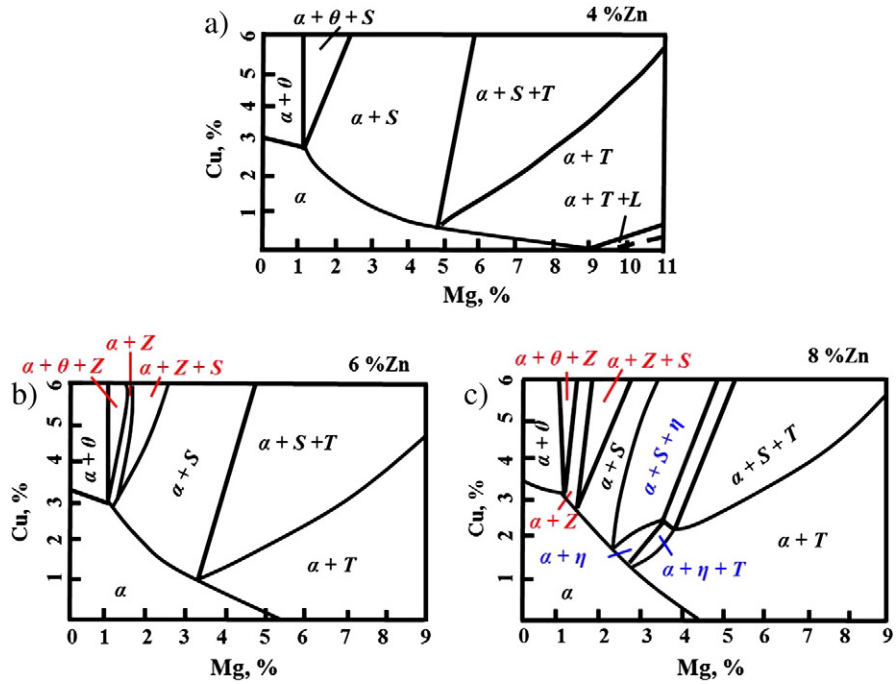

Fig. 18 – The isothermal cross section of Al–Zn–Mg–Cu phase diagram at 460 °C near the Al area [33].

limitation of Zn in Al matrix [25]. It should be noted that the composition of $Mg(Zn,Cu,Al)_2$ phase in the present work (Fig. 9) was slightly affected by the homogenization process. The dissolved $Mg(Zn,Cu,Al)_2$ phase during homogenization would lead to the increase of Zn content in Al matrix, as shown in Fig. 17. Therefore, the concentration gradient of Zn from $Mg(Zn,Cu,Al)_2$ phase to Al matrix would be reduced. The decrease of concentration gradient was more serious for the Al–Zn–Mg–Cu alloy with high Zn content, which was detrimental for the diffusion of Zn and phase transition from $Mg(Zn,Al,Cu)_2$ to S phase.

Moreover, the isothermal cross section of Al–Zn–Mg–Cu phase diagram at 460 °C near the Al area was shown in Fig. 18. It was obvious that in the alloys with 4% Zn content, the order of the phase regions was α + θ, α + θ + S, α + S, α + S + T, α + T and α + T + liquid phase with the increase of Mg content. When the Zn content increased to 6%, α + T phase expanded while α + S phase shrank. New phase Z (Mg_2Zn_{11}) also occurred. Therefore new phase regions α + θ + Z, α + Z and α + Z + S were detected between α + θ region and α + S region.

With Zn content increasing to 8%, α + T region and α + Z region further expanded. New regions α + η + T, α + η and α + S + η appeared between α + T region and α + S region, yet α + S region further decreased. Therefore, with the increase of Zn content, α + S region continuously decreased. The transformation from T phase to S phase turned to be harder during the homogenization process of Al–Zn–Mg–Cu alloy.

4. Conclusions

In the present work, the microstructure evolution of Al–8.35Zn–2.5Mg–2.25Cu alloy during homogenization treatments was investigated. The main phases of Al–Zn–Mg–Cu alloy before and after homogenization were α-Al, $Mg(Zn,Cu,Al)_2$, which had the structure similar to $MgZn_2$ containing Al and Cu elements, and Al_7Cu_2Fe phases. Only one endothermic peak at 747 K in DSC test, which is corresponding to the melting of $Mg(Zn,Cu,Al)_2$ phase, was observed regardless of homogenization time. According to DSC result and theoretical calculation result, 743 K was chosen as the homogenization temperature in the present work. The content of $Mg(Zn,Cu,Al)_2$ phase firstly decreased sharply within the initial 1 h, and then the mesh-like $Mg(Zn,Cu,Al)_2$ structure was gradually evolved to be discontinuous particles with the prolonging of homogenization time. The ratio of Al, Zn, Mg and Cu elements in $Mg(Zn,Cu,Al)_2$ phase was slightly affected by the homogenization treatment. Combining with other literature results, it could be concluded that the phase transition from $Mg(Zn,Al,Cu)_2$ to Al_2CuMg phases was very difficult when Zn content was higher than 8 wt.%. It is suggested that the significant effect of Zn might be explained by the diffusion behavior of Zn element. However, the mechanism should be further deeply investigated.

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